Register Number | | | | | | | | | |

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be University U/S 3 of UGC Act,1956)

Programme & Branch: B.E / B.Tech - CIVIL / EEE / Mechatronics / CSE with DS / CSE WITH CS / CS & BS / AI & DS / ECE / AERO / MECH / CSE with AI / CSE WITH AI & ML / ECE & DS/ CSE WITH AI & ROB

Course Title: Physics — Max. Marks: 100
Course Code : SPHB1104 (2025) — Time : 3 Hours
Date : 11-12-2025 — Session : FN

(NOTE: Assume relevant data, if found missing)

PART - A (10 × 2 = 20)

Answer ALL the Questions

1. State Hooke's law and its implications. (CO1)
2. Define torque and couple. (CO1)
3. List two types of point defects. (CO2)
4. What is a unit cell of the crystal? (CO2)
5. Prove the relation between and magnetic susceptibility and permeability. (CO3)
6. Draw and explain avalanche breakdown. (CO3)
7. Calculate De Broglie wavelength of a proton moving with a speed of 3×10^5 m/s. (CO4)
8. Write the time-independent Schrödinger equation. (CO4)
9. Mention any the four characteristics of laser. (CO5)
10. How cavity is used in LASER? (CO5)

PART - B (5 × 16 = 80)

Answer ALL the Questions

11. Describe the torsion pendulum experiment to find the expression for the rigidity modulus by using torsional oscillations. (CO1)

(or)

12. Explain the principle of uniform bending of a beam. Describe the experimental setup and procedure to determine Young's modulus using uniform bending. (CO1)

13. Compare the packing factor and atomic arrangements of SC, BCC, and FCC structures. (CO2)

(or)

14. Describe the Bridgeman technique for single crystal growth. Discuss its advantages and limitations. (CO2)

15. Discuss p-n junction diode. Explain its V-I characteristics under forward and reverse bias. (CO3)

(or)

16. Explain the concept of Fermi energy level in intrinsic and extrinsic semiconductors and how it shifts with doping. (CO3)

17. Derive the energy of an electron at n^{th} state and Calculate the energy at three consecutive states from ground state by using the derived expression. (CO4)

(or)

18. Explain the time-dependent Schrödinger wave equation. Derive the equation and discuss its physical significance. (CO4)

19. Compare Quantum Cascade Laser (QCL) and ILD in terms of working principle and construction. Draw energy level diagrams. (CO5)

(or)

20. How does a laser work? Describe the functions of the laser medium, population inversion, pumping process and optical resonator in laser operation. (CO5)